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COOLANT FLOW RATE CONTROL METHOD AND SERVER CABINET

Abstract

A coolant flow rate control method, configured to be applied to a plurality of servers and a fluid driver in fluid communication with the plurality of servers. The coolant flow rate control method includes setting a predetermined pressure difference between inlets and outlets of the plurality of servers based on power data of the plurality of servers and adjusting a duty ratio of the fluid driver for maintaining an actual pressure difference between the inlets and the outlets of the plurality of servers to match the predetermined pressure difference.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This non-provisional application claims priority under 35 U.S.C. § 119(a) on Patent Application No(s). 113108708 filed in Taiwan, R.O.C. on Mar. 8, 2024, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD

[0002] The disclosure relates to a coolant flow rate control method and a server cabinet.

BACKGROUND

[0003] Currently, servers in a cabinet are cooled by liquid cooling means. For example, in addition to a plurality of servers, there may be a fluid driver disposed in the cabinet. The fluid driver can drive a coolant to pass through the servers and carry away heat generated by heat sources (e.g. central processing units) in the servers. The existing fluid driver drives the coolant at a constant flow rate.

[0004] In general, the servers in the cabinet may be removed from the cabinet for maintenance. As the number of servers in the cabinet decreases, the flow rate of the coolant passing through each server may increase, which overly cools the remaining servers and increases an overall impedance. Therefore, in order to keep the coolant flowing at the set flow rate, the fluid driver needs to operate at a higher RPM (revolutions per minute), thereby increasing the power consumption.

SUMMARY

[0005] One embodiment of this disclosure provides a coolant flow rate control method, configured to be applied to a plurality of servers and a fluid driver in fluid communication with the servers. The coolant flow rate control method includes: setting a predetermined pressure difference between inlets and outlets of the servers based on power data of the servers, and adjusting a duty ratio of the fluid driver for maintaining an actual pressure difference between the inlets and the outlets of the servers to match the predetermined pressure difference.

[0006] Another embodiment of this disclosure provides a server cabinet. The server cabinet includes a plurality of servers, a fluid driver, and a main controller. The servers are connected in parallel to each other, and each of the servers has an inlet and an outlet. The fluid driver is in fluid communication with the servers. The main controller is electrically connected to the fluid driver. The main controller is configured to set a predetermined pressure difference between the inlets and the outlets of the servers based on power data of the servers, and adjust a duty ratio of the fluid driver for maintaining an actual pressure difference between the inlets and the outlets of the servers to match the predetermined pressure difference.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The present disclosure will become better understood from the detailed description given herein below and the accompanying drawings which are given by way of illustration only and thus are not intending to limit the present disclosure and wherein:

[0008] FIG. **1** is a schematic diagram of a server cabinet according to a first embodiment of the disclosure;

[0009] FIG. **2** is a flowchart of a coolant flow rate control method cooperated with the server cabinet in FIG. **1**;

[0010] FIG. **3** is a schematic diagram of the server cabinet in FIG. **1** when one of the servers is removed;

[0011] FIG. **4** is a schematic diagram of a server cabinet according to a second embodiment of the

disclosure;
[0012] FIG. **5** is a flowchart of a coolant flow rate control method cooperated with the server cabinet in FIG. **4**;
[0013] FIG. **6** is a schematic diagram of a server cabinet according to a third embodiment of the disclosure; and
[0014] FIG. **7** is a flowchart of a coolant flow rate control method cooperated with the server cabinet in FIG. **6**.
DETAILED DESCRIPTION
[0015] Firstly, note that, in the drawings, a thin line connects two components, which indicates that the two components are electrically connected to each other; that is, signals can be transmitted between them. Moreover, a thick arrow connects two components, which indicates that the two components are in fluid communication with each other and illustrates the flowing direction of fluid between them.
[0016] Referring to FIG. **1**. FIG. **1** is a schematic diagram of a server cabinet according to a first embodiment of the disclosure.
[0017] In this embodiment, the server cabinet **1** includes a plurality of servers **10**, a fluid driver **20**, and a main controller **30**. In addition, the server cabinet **1** may include a cabinet. The server **10**, the fluid driver **20**, and the main controller **30** are disposed in the cabinet. In order to clearly show the connection relationship among the server **10**, the fluid driver **20**, and the main controller **30**, the cabinet is omitted in FIG. **1**.
[0018] These servers **10** are connected in parallel to each other, and the fluid driver **20** is in fluid communication with these servers **10**. Furthermore, server cabinet **1** may include a first manifold **40** and a second manifold **50**. Each of these servers **10** has an inlet **11** and an outlet **12**. In addition, the fluid driver **20** also has an inlet **21** and outlet **22**. The outlet **22** of the fluid driver **20** is in fluid communication with the inlets **11** of these servers **10** through the first manifold **40**, and the inlet **21** of the fluid driver **20** is in fluid communication with the outlets **12** of these servers **10** through the second manifold **50**.
[0019] In this embodiment, the server cabinet **1** may further include a first pressure sensor **60** and a second pressure sensor **70**. The first pressure sensor **60** and the second pressure sensor **70** are configured to measure the pressures of the inlets **11** and the outlets **12** of the servers **10**, respectively. For example, the first pressure sensor **60** is disposed in the first manifold **40** and located close to the top of the first manifold **40**, and the second pressure sensor **70** is disposed in the second manifold **50** and located close to the top of the second manifold **50**.
[0020] The main controller **30** is electrically connected to the fluid driver **20**, the first pressure sensor **60**, and the second pressure sensor **70**.
[0021] The following paragraphs will introduce a coolant flow rate control method cooperated with the server cabinet **1**. Referring to FIG. **1** and FIG. **2**. FIG. **2** is a flowchart of a coolant flow rate control method that could cooperate with the server cabinet in FIG. **1**.
[0022] First, the step of setting a predetermined pressure difference between the inlets **11** and the outlets **12** of the servers **10** based on power data of the servers is performed. For example, the step of setting a predetermined pressure difference between the inlets **11** and the outlets **12** of the servers **10** based on power data of the servers includes a step S**01**. The step S**01** is to set the predetermined pressure difference between the inlets **11** and the outlets **12** of the servers **10** based on maximum operation power data of the servers **10**, where the predetermined pressure difference is a maximum predetermined pressure difference.
[0023] Furthermore, the main controller **30** of this embodiment is not electrically connected to the servers **10**. The predetermined pressure difference between the inlets **11** and the outlets **12** of the servers **10** set by the main controller **30** is based on, for example, the maximum operation power data of the servers **10** at a full load state which are manually given, and the predetermined pressure difference is a maximum predetermined pressure difference. Assuming that the cabinet

accommodates three servers **10**, a heat source **13** (e.g., CPU or graphic processing unit, GPU) in each server **10** operates in a maximum power, the predetermined pressure difference is set to 100 Kpa, the duty ratio of the fluid driver **20** is 100%, and a total flow rate of the coolant output by the fluid driver **20** is 100 LPM (liter per minute), the flow rate of the coolant through each server **10** is 33.3 LPM, and the heat source **13** in each server **10** can operate in an appropriate temperature.

[0024] Next, a Step S**03** is performed to adjust a duty ratio of the fluid driver **20** for maintaining an actual pressure difference between the inlets **11** and the outlets **12** of the servers **10** to match the predetermined pressure difference. For example, the main controller **30** adjusts the duty ratio of the fluid driver **20** to match the actual pressure difference between the inlets **11** and the outlets **12** of the servers **10** to the predetermined pressure difference. The actual pressure difference is the difference between the pressures measured by the first pressure sensor **60** and the second pressure sensor **70**. The actual pressure difference matches the predetermined pressure difference, which means that the actual pressure difference is equal to the predetermined pressure difference, or the actual pressure difference falls within a range of the predetermined pressure difference plus and minus a predetermined value.

[0025] For further illustration, referring to FIG. **3**. FIG. **3** is a schematic diagram of the server cabinet in FIG. **1** when one of the servers is removed. Assuming that one of the servers **10** is removed from the cabinet for maintenance, the actual pressure difference between the inlets **11** and the outlets **12** of the remaining servers **10** will inevitably be changed and may not match the predetermined pressure difference. In order to restore the actual pressure difference between the inlets **11** and the outlets **12** of these servers **10** to match the predetermined pressure difference, the main controller **30** will adjust the duty ratio of the fluid driver **20** for maintaining the actual pressure difference between the inlets **11** and the outlets **12** of these servers **10** to match the predetermined pressure difference.

[0026] Referring to Table 1. Table 1 shows a comparison between the constant pressure difference control manner of this embodiment and a conventional constant flow rate control manner when the number of the servers **10** is reduced. The constant pressure difference control manner of this embodiment is the aforementioned step of maintaining the actual pressure difference between the inlets **11** and the outlets **12** of these servers **10** to match the predetermined pressure difference, and the conventional constant flow rate control manner refers to maintain the total flow rate of the coolant output by the fluid driver **20**.

[0027] In Table 1, Scenario A is a reference, the data in Scenario A are expressed as 100%, and the data in Scenario B are percentage values compared to the data in Scenario A.

TABLE-US-00001 TABLE 1 Scenario A: Connecting three servers B: Connecting two servers Constant Constant pressure Constant flow pressure Constant flow Control manner difference rate difference rate Pressure difference 100% 100% 150% between inlets and outlets of servers Total flow rate of 100% 66% 100% coolant output from fluid driver Duty ratio of fluid 100% 80% 115% driver

[0028] As shown in Table 1, in the constant pressure difference control manner of this embodiment, the main controller **30** reduces the duty ratio of the fluid driver **20**, for example, from 100% to 80%, such that the total flow rate of the coolant output from the fluid driver **20**, for example, decreases from 100% to 66% for restoring the actual pressure difference between the inlets **11** and the outlets **12** of these servers **10** to match the predetermined pressure difference. As for the conventional constant flow rate control manner, the pressure difference between the inlets and the outlets of the servers increases to 150%, and the duty ratio of the fluid driver **20** increases to 115%.

[0029] As a result, when the number of servers **10** is reduced due to maintenance, the flow rate of the coolant output by the fluid driver **20** can be reduced by decreasing the duty ratio of the fluid driver **20** for matching the actual pressure difference between the inlets **11** and outlets **12** of those servers **10** and the predetermined pressure difference, such that the power consumption of the fluid driver **20** can be reduced compared with the conventional constant flow rate control manner,

thereby saving energy.
[0030] After the step S**03** is finished, the step S**01** is performed again.
[0031] Then, referring to FIG. **4**, FIG. **4** is a schematic diagram of a server cabinet according to a second embodiment of the disclosure.
[0032] The server cabinet **1***a* of this embodiment is similar to the server cabinet **1** of the previous embodiment. The following mainly describes the differences between them, and the same parts can refer to the descriptions in the above paragraphs and will not be repeatedly introduced hereinafter.
[0033] In this embodiment, each of these servers **10** has a proportional valve **14** disposed at the inlet **11**, a temperature sensor **15** and a baseboard management controller **16**. The temperature sensor **15** is configured to measure the temperature of the heat source **13**, and the baseboard management controller **16** is electrically connected to the proportional valve **14**, the heat source **13**, and the temperature sensor **15**.
[0034] Next, referring to FIGS. **4** and **5**, FIG. **5** is a flowchart of a coolant flow rate control method that could cooperate with the server cabinet in FIG. **4**.
[0035] In addition to the above steps S**01** and S**03**, the coolant flow rate control method further includes steps S**21** to S**23**, which may be arranged between the steps S**01** and S**03**. In other words, after the step S**01** is performed to set the predetermined pressure difference between the inlets **11** and the outlets **12** of the servers **10** based on maximum operation power data of the servers **10**, where the predetermined pressure difference is a maximum predetermined pressure difference, the step S**21** is performed to determine whether temperatures of the heat sources **13** of these servers **10** are smaller than a predetermined temperature. When the temperature of the heat source **13** of at least one of these servers **10** is smaller than the predetermined temperature, the step S**22** is performed to reduce an opening degree of the proportional valve **14** at the inlet **11** of the at least one of these servers **10**, and then the step S**03** is performed to adjust the duty ratio of the fluid driver **20** for maintaining the actual pressure difference between the inlets **11** and the outlets **12** of these servers **10** to match the predetermined pressure difference.
[0036] For example, assuming that the number of these servers **10** is three, the heat source **13** (e.g., CPU or GPU) in each server **10** operates in a maximum power, the opening degree of the proportional valve **14** in each server **10** is 100%, the predetermined pressure difference is set to 100 Kpa, the duty ratio of the fluid driver **20** is 100%, and the total flow rate of coolant output from the fluid driver **20** is 100 LPM, the flow rate of coolant through each server **10** is 33.3 LPM, and the heat source **13** in each server **10** can operate in an appropriate temperature.
[0037] In a case that two of the servers **10** change from the full load state to a non-full load state (e.g., an idle state), the powers of the heat sources **13** of the two servers **10** are reduced, for example, a half, and the temperatures of the heat sources **13** are thereby reduced to, for example, about 80% of the original temperature. Once the baseboard management controllers **16** of the two servers **10** determine that the temperatures of the heat sources **13** transmitted by the temperature sensors **15** are smaller than the predetermined temperature, it indicates that the heat sources **13** of the two servers **10** are overly cooled by the coolant. At this moment, the baseboard management controllers **16** of the two servers **10** adjust the opening degrees of the proportional valves **14** (e.g., adjust the opening degrees of the proportional valves **14** to 50%). As for the other server **10**, the temperature of the heat source **13** of this server **10** is substantially maintained to be its original temperature because the heat source **13** of this server **10** is still at the full load state. Therefore, the baseboard management controller **16** of this server **10** still maintains the opening degree of the proportional valve **14** to be 100%.
[0038] After the opening degrees of the proportional valves **14** of the two servers **10** are adjusted, the actual pressure difference between the inlets **11** and the outlets **12** of these servers **10** will change and may not match the predetermined pressure difference. When the main controller **30** determines that the actual pressure difference between the inlets **11** and the outlets **12** of these servers **10** does not match the predetermined pressure difference by the pressures transmitted from

the first pressure sensor **60** and second pressure sensor **70**, the main controller **30** will adjust the duty ratio of the fluid driver **20** to restore the actual pressure difference between the inlets **11** and the outlets **12** of these servers **10** to match the predetermined pressure difference.

[0039] In detail, the main controller **30** will reduce the duty ratio of the fluid driver **20**, for example, from 100% to 80%, such that the total flow rate of the coolant output from the fluid driver **20** is reduced, for example, from 100 LPM to 66 LPM. At this moment, since the opening degrees of the proportional valves **14** of two of the three servers **10** are, for example, 50%, and the opening degree of the proportional valve **14** of the other server **10** is, for example, 100%, the flow rate of the coolant through the two servers **10** with the proportional valves **14** of 50% opening degree, for example, 16.7 LPM, and the flow rate of the coolant through the server **10** with the proportional valve **14** of 100% opening degree is, for example, 33.3 LPM. As a result, the actual pressure difference between the inlets **11** and the outlets **12** of these servers **10** is restored to match the predetermined pressure difference.

[0040] In another example, when all servers **10** change from the full load state to the non-full load state, the temperatures of the heat sources **13** of these servers **10** will decrease. Once the baseboard management controllers **16** of these servers **10** determines that the temperatures of the heat sources **13** transmitted by the temperature sensors **15** are smaller than the predetermined temperature, it means that the heat sources **13** of these servers **10** are overly cooled by the coolant. At this moment, the baseboard management controllers **16** of these servers **10** adjust the opening degrees of the proportional valves **14** (e.g., adjusting opening degrees of the proportional valves **14** to 50%).

[0041] After the opening degrees of the proportional valves **14** of these servers **10** are adjusted, the actual pressure difference between the inlets **11** and the outlets **12** of these servers **10** will be changed and may not match the predetermined pressure difference. When the main controller **30** determines that the actual pressure difference between the inlets **11** and the outlets **12** of these servers **10** does not match the predetermined pressure difference by the pressures transmitted from the first pressure sensor **60** and the second pressure sensor **70**, the main controller **30** will adjust the duty ratio of the fluid driver **20** to restore the actual pressure difference between the inlets **11** and the outlets **12** of these servers **10** to match the predetermined pressure difference.

[0042] In detail, the main controller **30** will reduce the duty ratio of the fluid driver **20**, for example, from 100% to 75%, such that the total flow rate of the coolant output from the fluid driver **20** is reduced, for example, from 100 LPM to 50 LPM. At this moment, since the opening degree of the proportional valve **14** of each of these servers **10** is, for example, 50%, the flow rate of the coolant flowing through each of the servers **10** is, for example, 16.7 LPM. As a result, the actual pressure difference between the inlets **11** and the outlets **12** of these servers **10** is restored to match the predetermined pressure difference.

[0043] Referring to Table 2, Table 2 shows the comparison between the constant pressure difference control manner with regulating the opening degrees of the proportional valves in this embodiment and the conventional constant flow rate control manner when the loadings of the servers **10** change.

[0044] In Table 2, Scenario A is a reference, the data in Scenario A are expressed as 100%, and the data in Scenarios B and C are percentage values compared to the data in Scenario A.

TABLE-US-00002 TABLE 2 Scenario A: All servers are B: Two of servers are C: All servers are at full load not at full load not at full load Constant Constant Constant pressure pressure pressure difference difference difference control control control manner with manner with manner with regulating regulating regulating opening opening opening degrees of Constant degrees of Constant degrees of Constant Control proportional flow proportional flow proportional flow manner valves rate valves rate valves rate Pressure 100% 100% 100% 100% 100% difference between inlets and outlets of servers Total flow 100% 66% 100% 50% 100% rate of coolant output from fluid driver Duty ratio 100% 80% 100% 75% 100% of fluid driver

[0045] As shown in Table 2, in Scenario B of the constant pressure difference control manner of this embodiment with regulating the opening degrees of the proportional valves, the main controller **30** reduces the duty ratio of the fluid driver **20**, for example, from 100% to 80%, such that the total flow rate of the coolant output from the fluid driver **20**, for example, decreases from 100% to 66% for restoring the actual pressure difference between the inlets **11** and the outlets **12** of these servers **10** to match the predetermined pressure difference. In Scenario C, the main controller **30** reduces the duty ratio of the fluid driver **20**, for example, from 100% to 75%, such that the total flow rate of the coolant output from the fluid driver **20** decreases, for example, from 100% to 50% for restoring the actual pressure difference between the inlets **11** and the outlets **12** of these servers **10** to match the predetermined pressure difference. In the conventional constant flow rate control manner, even if at least one of the servers is in the non-full load state, the duty ratio of the fluid driver **20** does not change since the total flow rate of the coolant is constant. As a result, the coolant flow rate control method of this embodiment allows the duty ratio of the fluid driver **20** to reduce to, for example, 80% and 75% in scenarios B and C, respectively, thereby reducing about 20% of the power consumption of the fluid driver **20**. Accordingly, in a case that the power of the heat source **13** of at least one of the servers **10** decreases, the opening degree of the proportional valve **14** of this server **10** can be adjusted by determining that the temperature of the heat source **13** of this server **10** is smaller than the predetermined temperature, and the actual pressure difference between the inlets **11** and the outlets **12** of the servers **10** can be maintained to match to the predetermined pressure difference, such that the power consumption of the fluid driver **20** can be reduced compared with the conventional constant flow rate control manner, thereby saving energy.

[0046] In the step S**21**, if the temperatures of the heat sources **13** of these servers **10** are not smaller than the predetermined temperature, the step S**23** is performed to maintain the opening degrees of the proportional valves **14** at the inlets **11** of these servers **10**. After the step S**23** is finished, the step S**03** is performed.

[0047] It should be noted that the step S**21** is not limited to be performed after the step S**01**. In other embodiments, the step S**21** and the step S**01** may be performed simultaneously, or the step S**21** may be performed before the step S**01**.

[0048] Next, referring to FIG. **6**, FIG. **6** is a schematic diagram of a server cabinet according to a third embodiment of the disclosure.

[0049] The server cabinet **1***b* of this embodiment is similar to the server cabinet **1** of the previous embodiment, and the following mainly describes the differences between them, and the same parts can refer to the descriptions in the above paragraphs and will not be repeatedly introduced hereinafter.

[0050] In this embodiment, the main controller **30** is electrically connected to these servers **10** and can obtain current power data for these heat sources **13** of these servers **10**.

[0051] Next, referring to FIGS. **6** and **7**, FIG. **7** is a flowchart of a coolant flow rate control method cooperated with the server cabinet in FIG. **6**.

[0052] In the coolant flow rate control method, the first step is to set a predetermined pressure difference between the inlets **11** and the outlets **12** of these servers **10** based on power data of these servers **10**, where the aforementioned step includes setting the predetermined pressure difference between the inlets **11** and the outlets **12** of these servers **10** based on the current power data of these servers **10**. For example, the aforementioned step includes step S**31**: determining whether all of the current power data of these servers **10** are smaller than maximum operation power data. If yes, a step S**32** is performed to set the predetermined pressure difference between the inlets **11** and the outlets **12** of these servers **10** to be smaller than a maximum predetermined pressure difference.

[0053] In detail, the maximum operation power data are set to the powers of these servers **10** when they operate in the full load state. When the main controller **30** determines that the current power data of all of these servers **10** are smaller than the maximum operation power data, it can be understood that all of these servers **10** are not in the full load state. Thus, the main controller **30** does not require to set the predetermined pressure difference to the maximum predetermined pressure difference, but set the predetermined pressure difference to be smaller than the maximum predetermined pressure difference.

[0054] After the step S**32**, a step S**33** is performed to determine whether all of the current power data for these servers **10** are equal to one another. If yes, a step S**34** is performed to fully open the proportional valves **14** at the inlets **11** of these servers **10**. Next, a step S**35** is performed to adjust the duty ratio of the fluid driver **20** for maintaining the actual pressure difference between the inlets **11** and the outlets **12** of these servers **10** to match the predetermined pressure difference.

[0055] Taking an example to illustrate the steps S**31** to S**35**, assuming that the number of the servers **10** is three, the heat source **13** in each server **10** operates in a maximum power, the opening degree of the proportional valve **14** in each server **10** is 100%, the predetermined pressure difference is set to the maximum predetermined pressure difference (e.g., 100 Kpa), the duty ratio of the fluid driver **20** is set to 100%, and the total flow rate of the coolant output from the fluid driver **20** is 100 LPM, the flow rate of the coolant through each server **10** is 33.3 LPM, and the heat sources **13** in the servers **10** can operate in an appropriate temperature. In a case that the heat sources **13** in the servers **10** change to operate in same powers smaller than the maximum power (i.e., the current power data of these servers **10** are equal to each other and are smaller than the maximum operation power data), the main controller **30** sets the predetermined pressure difference (e.g., 42 Kpa) between the inlets **11** and the outlets **12** of these servers **10** to be smaller than the maximum predetermined pressure difference (e.g., 100 Kpa), and the main controller **30** fully open the proportional valves **14** via the baseboard management controllers **16** of these servers **10**. In order to maintain the actual pressure difference between the inlets **11** and the outlets **12** of these servers **10** to match the predetermined pressure difference, the duty ratio of the fluid driver **20** is adjusted to be, for example, 60% by the main controller **30**, and the total flow rate of the coolant output from the fluid driver **20** is, for example, 50 LPM. At this moment, since the opening degrees of the proportional valves **14** of these servers **10** are equal to one another, the flow rate of the coolant through each server **10** is, for example, 16.7 LPM.

[0056] Referring to Table 3, Table 3 shows the comparison between the constant pressure difference control manner after adjusting pressure difference in this embodiment and the conventional constant flow rate control manner when the power load of the server **10** is changed. In Table 3, Scenario A is a reference, the data in Scenario A are expressed as 100%, and the data in Scenario B are percentage values compared to the data in Scenario A.

TABLE-US-00003 TABLE 3 Scenario A: All servers are at full load B: All servers are not at full load Constant Constant pressure pressure difference difference control manner control manner after adjusting after adjusting pressure Constant flow pressure Constant flow Control manner difference rate difference rate Pressure 100% 42% 100% difference between inlets and outlets of servers Total coolant 100% 50% 100% flow output from fluid driver Duty ratio of 100% 60% 100% fluid driver

[0057] As shown in Table 3, in Scenario B of the constant pressure difference control manner after adjusting pressure difference in this embodiment, the main controller **30** reduces the predetermined pressure difference between the inlets **11** and the outlets **12** of these servers **10** to, for example, 42%, the main controller **30** reduces the duty ratio of the fluid driver **20** from, for example, 100% to 60%, such that the total flow rate of the coolant output from the fluid driver **20** decreases from, for example, 100% to 50% for matching the actual pressure difference between the inlets **11** and the outlets **12** of these servers **10** and the reduced predetermined pressure difference. In the conventional constant flow rate control manner, even if the servers are not in the full load state, the duty ratio of the fluid driver **20** does not change since the total flow rate of the coolant is constant. As a result, in a case that the main controller **30** determines that the heat sources **13** of all the servers **10** are not in the full load state, setting the predetermined pressure difference between the inlets **11** and the outlets **12** of these servers **10** to be less than the maximum predetermined pressure difference and making the actual pressure difference between the inlets **11** and the outlets **12** of these servers **10** to match the predetermined pressure difference can reduce the power consumption

of the fluid driver **20** compared with the conventional constant flow rate control manner, thereby saving energy.
[0058] In step S**31**, when the current power data of these servers **10** are determined to be not all smaller than the maximum operation power data, a step S**36** is performed to set the predetermined pressure difference between the inlets **11** and the outlets **12** of these servers **10** to be equal to the maximum predetermined pressure difference. Then, a step S**37** is performed to determine whether the temperatures of the heat sources **13** of the servers **10** are smaller than the predetermined temperature.
[0059] In step S**33**, when the current power data of the servers **10** are not equal to one another, the step S**37** is performed determine whether the temperatures of the heat sources **13** of the servers **10** are smaller than the predetermined temperature.
[0060] In step S**37**, When the temperature of the heat source **13** of at least one of the servers **10** is smaller than the predetermined temperature, a step S**38** is performed to reduce the opening degree of the proportional valve **14** at the inlet **11** of the at least one of the servers **10**. Then, the step S**35** is performed to adjust the duty ratio of the fluid driver **20** for maintaining the actual pressure difference between the inlets **11** and the outlets **12** of the servers **10** to match the predetermined pressure difference.
[0061] In step S**37**, when the temperatures of the heat sources **13** of the servers **10** are not smaller than the predetermined temperature, a step S**39** is performed to maintain the opening degrees of the proportional valves **14** at the inlets **11** of the servers **10**. After step S**39** is finished, the step S**35** is performed.
[0062] The aforementioned steps S**36**, S**37**, S**38**, and S**39** are the same as the steps S**01**, S**21**, S**22**, and S**23**, and thus the details of the steps S**36**, S**37**, S**38**, and S**39** will not be described repeatedly and can refer to the descriptions in the above paragraphs.
[0063] Note that step S**32** is not limited to be performed when the current power data of these servers **10** are all smaller than the maximum operation power data. In other embodiments, the step S**32** may be performed when at least part of the current power data of these servers **10** is smaller than the maximum operation power data. In such a case, the step S**36** will be performed when the current power data of these servers **10** are all not smaller than the maximum operation power data.
[0064] In addition, the step S**34** is not limited to fully opening the proportional valves **14** at the inlets **11** of the servers **10**. In other embodiments, the step S**34** may be modified to set the opening degrees of the proportional valves at the inlets of these servers to a predetermined opening degree, where the predetermined opening degree may be any value from 1% to 100%.
[0065] Furthermore, the step S**33** and the step S**34** are optional steps. In other embodiments, the step S**33** and the step S**34** may be omitted, and after the step S**32** is performed, the step S**37** may be performed directly. In addition, the step S**37** is not limited to being performed after the step S**31** is finished. In other embodiments, the step S**37** and the step S**31** may be performed simultaneously, or the step S**37** may be performed before the step S**31**.
[0066] According to the coolant flow rate control method and server cabinet disclosed in the above embodiments, the flow rate of the coolant output by the fluid driver can be reduced by adjusting the duty ratio of the fluid driver for matching the actual pressure difference between the inlets and the outlets of the servers with the predetermined pressure difference, which reduces the power consumption of the fluid driver, thereby saving energy.
[0067] In addition, in a case that the power of the heat source of at least part of the servers decreases, the opening degree of the proportional valve of this server can be adjusted by determining that the temperature of the heat source of this server is smaller than the predetermined temperature, and the actual pressure difference between the inlets and the outlets of these servers can be maintained to match to the predetermined pressure difference, such that the power consumption of the fluid driver can be reduced compared with the conventional constant flow rate control manner, thereby saving energy.

[0068] In a case that the main controller determines that the heat sources of all servers are not in the full load, setting the predetermined pressure difference between the inlets and the outlets of these servers to be less than the maximum predetermined pressure difference, and making the actual pressure difference between the inlets and the outlets of these servers to match the predetermined pressure difference can reduce the power consumption of the fluid driver compared with the conventional constant flow rate control manner, thereby saving energy.
[0069] It will be apparent to those skilled in the art that various modifications and variations can be made to the present disclosure. It is intended that the specification and examples be considered as exemplary embodiments only, with a scope of the disclosure being indicated by the following claims and their equivalents.

Claims

1. A coolant flow rate control method, configured to be applied to a plurality of servers and a fluid driver in fluid communication with the plurality of servers, comprising: setting a predetermined pressure difference between inlets and outlets of the plurality of servers based on power data of the plurality of servers; and adjusting a duty ratio of the fluid driver for maintaining an actual pressure difference between the inlets and the outlets of the plurality of servers to match the predetermined pressure difference.
2. The coolant flow rate control method according to claim 1, wherein the step of setting the predetermined pressure difference between the inlets and the outlets of the plurality of servers based on the power data of the plurality of servers comprises setting the predetermined pressure difference between the inlets and the outlets of the plurality of servers based on maximum operation power data of the plurality of servers, and the predetermined pressure difference is a maximum predetermined pressure difference.
3. The coolant flow rate control method according to claim 2, further comprising: determining whether temperatures of heat sources of the plurality of servers are smaller than a predetermined temperature; if the temperature of the heat source of at least one of the plurality of servers is smaller than the predetermined temperature, reducing an opening degree of a proportional valve at the inlet of the at least one of the plurality of servers, and performing the step of adjusting the duty ratio of the fluid driver for maintaining the actual pressure difference between the inlets and the outlets of the plurality of servers to match the predetermined pressure difference.
4. The coolant flow rate control method according to claim 3, wherein the step of determining whether the temperatures of the heat sources of the plurality of servers are smaller than the predetermined temperature is performed after the step of setting the predetermined pressure difference between the inlets and the outlets of the plurality of servers based on the maximum operation power data of the plurality of servers.
5. The coolant flow rate control method according to claim 1, wherein the step of setting the predetermined pressure difference between the inlets and the outlets of the plurality of servers based on the power data of the plurality of servers comprises setting the predetermined pressure difference between the inlets and the outlets of the plurality of servers based on current power data of the plurality of servers, and comprises: determining whether at least part of the current power data of the plurality of servers is smaller than maximum operation power data; if yes, setting the predetermined pressure difference to be smaller than a maximum predetermined pressure difference.
6. The coolant flow rate control method according to claim 5, wherein the step of determining whether at least part of the current power data of the plurality of servers is smaller than the maximum operation power data comprises determining whether all of the current power data of the plurality of servers are smaller than the maximum operation power data.
7. The coolant flow rate control method according to claim 6, wherein if the current power data of

the plurality of servers are not all smaller than the maximum operation power data, setting the predetermined pressure difference to be equal to the maximum predetermined pressure difference.

8. The coolant flow rate control method according to claim 5, further comprising: determining whether temperatures of heat sources of the plurality of servers are smaller than a predetermined temperature; if the temperature of the heat source of at least one of the plurality of servers is smaller than the predetermined temperature, reducing an opening degree of a proportional valve at the inlet of the at least one of the plurality of servers, and performing the step of adjusting the duty ratio of the fluid driver for maintaining the actual pressure difference between the inlets and the outlets of the plurality of servers to match the predetermined pressure difference.

9. The coolant flow rate control method according to claim 8, wherein the step of determining whether the temperatures of the heat sources of the plurality of servers are smaller than the predetermined temperature is performed after the step of setting the predetermined pressure difference between the inlets and the outlets of the plurality of servers based on the current power data of the plurality of servers.

10. The coolant flow rate control method according to claim 8, after the step of setting the predetermined pressure difference to be smaller than the maximum predetermined pressure difference, further comprising: determining whether the current power data of the plurality of servers are equal to one another; if yes, fully opening proportional valves at the inlets of the plurality of servers, and performing the step of adjusting the duty ratio of the fluid driver for maintaining the actual pressure difference between the inlets and the outlets of the plurality of servers match the predetermined pressure difference; and if not, performing the step of determining whether the temperatures of the heat sources of the plurality of servers are smaller than the predetermined temperature.

11. The coolant flow rate control method according to claim 1, wherein the power data of the plurality of servers comprises current power data of the plurality of servers or maximum operation power data of the plurality of servers.

12. A server cabinet, comprising: a plurality of servers, wherein the plurality of servers are connected in parallel to each other, and each of the plurality of servers has an inlet and an outlet; a fluid driver, in fluid communication with the plurality of servers; and a main controller, electrically connected to the fluid driver; wherein the main controller is configured to set a predetermined pressure difference between the inlets and the outlets of the plurality of servers based on power data of the plurality of servers and adjust a duty ratio of the fluid driver for maintaining an actual pressure difference between the inlets and the outlets of the plurality of servers to match the predetermined pressure difference.

13. The server cabinet according to claim 12, wherein the main controller is configured to set the predetermined pressure difference between the inlets and the outlets of the plurality of servers based on maximum operation power data of the plurality of servers, and the predetermined pressure difference is a maximum predetermined pressure difference.

14. The server cabinet according to claim 13, wherein each of the plurality of servers comprises a proportional valve disposed at the inlet, a heat source, a temperature sensor, and a baseboard management controller, the temperature sensor is configured to measure a temperature of the heat source, and the baseboard management controller is electrically connected to the proportional valve, the heat source, and the temperature sensor; when the baseboard management controller of at least one of the plurality of servers determines that the temperature of the heat source is smaller than a predetermined temperature, the baseboard management controller of the at least one of the plurality of servers reduces an opening degree of the proportional valve at the inlet, and the main controller adjusts the duty ratio of the fluid driver for maintaining the actual pressure difference between the inlets and the outlets of the plurality of servers to match the predetermined pressure difference.

15. The server cabinet according to claim 14, wherein after the main controller sets the predetermined pressure difference between the inlets and the outlets of the plurality of servers based on the maximum operation power data of the plurality of servers, the baseboard management controller of each of the plurality of servers determines whether the temperature of the heat source is smaller than the predetermined temperature.

16. The server cabinet according to claim 12, wherein the main controller is electrically connected to the plurality of servers, the main controller is configured to set the predetermined pressure difference between the inlets and the outlets of the plurality of servers based on current power data of the plurality of servers, and the main controller is configured to set the predetermined pressure difference to be smaller than a maximum predetermined pressure difference when at least part of the current power data is smaller than maximum operation power data.

17. The server cabinet according to claim 16, wherein the main controller is configured to set the predetermined pressure difference to be smaller than the maximum predetermined pressure difference when all of the current power data are smaller than the maximum operation power data, and the main controller is configured to set the predetermined pressure difference to be equal to the maximum predetermined pressure difference when the current power data are not all smaller than the maximum operation power data.

18. The server cabinet according to claim 16, wherein each of the plurality of servers comprise a proportional valve disposed at the inlet, a heat source, a temperature sensor, and a baseboard management controller, the temperature sensor is configured to measure a temperature of the heat source, and the baseboard management controller is electrically connected to the proportional valve, the heat source, and the temperature sensor; when the baseboard management controller of at least one of the plurality of servers determines that the temperature of the heat source is smaller than a predetermined temperature, the baseboard management controller of the at least one of the plurality of servers reduces an opening degree of the proportional valve at the inlet, and the main controller adjusts the duty ratio of the fluid driver for maintaining the actual pressure difference between the inlets and the outlets of the plurality of servers to match the predetermined pressure difference.

19. The server cabinet according to claim 18, wherein after the main controller sets the predetermined pressure difference between inlets and outlets of the plurality of servers based on the current power data of the plurality of servers, the baseboard management controller of each of the plurality of servers determines whether the temperature of the heat source is smaller than the predetermined temperature.

20. The server cabinet according to claim 18, wherein after the main controller sets the predetermined pressure difference to be smaller than the maximum predetermined pressure difference, the main controller is configured to fully open the proportional valve at the inlet via the baseboard management controller of each of the plurality of servers when all of the current power data of the plurality of servers are equal to one another, and the main controller is configured to drive the baseboard management controllers of the plurality of servers to determine whether the temperatures of the heat sources are smaller than the predetermined temperature when the current power data of the plurality of servers are not all equal to one another.

21. The server cabinet according to claim 12, further comprising a first pressure sensor and a second pressure sensor, wherein the first pressure sensor and the second pressure sensor are electrically connected to the main controller, and the first pressure sensor and the second pressure sensor are respectively configured to measure pressures of the inlets and the outlets of the plurality of servers, and the actual pressure difference is a difference between the pressures measured by the first pressure sensor and the second pressure sensor.

22. The server cabinet according to claim 21, further comprising a first manifold and a second manifold, wherein an outlet of the fluid driver is connected to the inlets of the plurality of servers via the first manifold, an inlet of the fluid driver is connected to the outlets of the plurality of servers via the second manifold, the first pressure sensor is disposed in the first manifold, and the second pressure sensor is disposed in the second manifold.

23. The server cabinet according to claim 12, wherein the power data of the plurality of servers comprises current power data of the plurality of servers or maximum operation power data of the plurality of servers.